

Diversifying Differentiable Graph Retrieval with Topic-Adaptive Multi-Intent Learning

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Abstract

Knowledge Graph Question Answering (KGQA) leverages structured knowledge graphs for reliable reasoning. Graph-based Retrieval-Augmented Generation (RAG) addresses the incompleteness and hallucination issues of Large Language Models (LLMs) by retrieving query-relevant subgraphs. However, existing approaches rely on single-intent semantic retrieval, compressing queries into single representations and optimizing each query independently. This leads to narrow triple selection that omits complementary information. While multi-intent retrieval diversification addresses this limitation, it faces critical challenges: (1) semantic diversification does not guarantee reasoning performance, and (2) co-selection frequency across intents does not ensure reasoning benefit. We propose **Topic-Adaptive Retrieval Diversification (TARD)**, based on end-to-end optimization via generation feedback. TARD adaptively extracts multiple topic-based intents through neural topic modeling and employs a Gumbel-Softmax differentiable sampling to enable joint optimization. Supervised fine-tuning aligns the topic-adaptive multi-intent selector and triple scorer with reasoning performance to achieve beneficial consensus patterns. Adaptive direct preference optimization trains the generator to utilize relevant consensus while ignoring uninformative patterns. Experiments on WebQSP and CWQ show that TARD outperforms state-of-the-art baselines. Our code and data are publicly available at <https://github.com/leedongcheon/TARD>.

CCS Concepts

• Information systems → Question answering.

Keywords

Knowledge Graph Question Answering, Retrieval-Augmented Generation, Multi-Intent Retrieval, End-to-end Optimization

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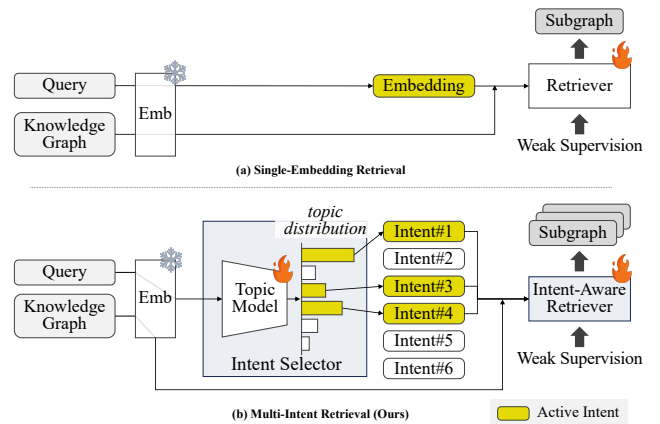


Figure 1: Comparison of graph retrieval paradigms for KGQA. (a) Existing single-embedding approach compresses each query into one representation. (b) Our multi-intent approach extracts diverse semantic facets through topic modeling.

1 Introduction

Knowledge Graph Question Answering (KGQA) aims to answer questions by reasoning over structured knowledge graphs that encode entities and relations with explicit semantics. Traditional KGQA approaches employed graph neural networks [27, 47] to propagate information along graph neighborhoods and embedding-based models [29, 43] to align queries with KG elements. These methods struggle with scalability and compositional query understanding. Recently, Large Language Models (LLMs) have been widely adopted for relation extraction across diverse NLP tasks [4, 7, 41], benefiting from improved query understanding and reasoning, while still suffering from incompleteness and hallucination. To address this, graph-based Retrieval-Augmented Generation (RAG) [13, 32, 36, 37] has emerged as a promising approach for

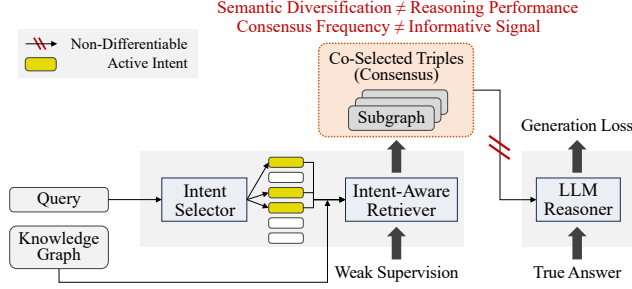


Figure 2: Challenges in naive multi-intent retrieval. The top-B hard selection operator is non-differentiable, preventing generation feedback from optimizing the retriever.

KGQA by incorporating external structured evidence. Compared to document-centric RAG [11, 21] that retrieves unstructured text, graph-based RAG leverages query-relevant subgraphs with explicit entity-relation chains, enabling traceable reasoning and factual reliability.

Despite these advantages, existing graph-based RAG approaches suffer from compressing each query into a single representation as shown in Figure 1(a). Real-world queries often contain multiple semantic facets requiring diverse retrieval perspectives. However, current retrievers encode queries as unified embeddings for retrieval. Recent methods employ various approaches including graph neural networks [28], dense vector representations [14, 45], semantic path matching [12], and subgraph retrieval with smaller language models [16]. Most recently, SubgraphRAG [20] achieves state-of-the-art performance through shortest path supervision. However, these approaches fundamentally compress multi-faceted queries into single embeddings, leading to narrow triple selection that omits complementary information within individual queries.

To address this limitation, we propose **Topic-Adaptive Retrieval Diversification (TARD)**, which leverages multiple intents extracted from neural topic modeling jointly optimized with the subgraph retriever. Unlike explicit decomposition methods [17, 19, 35, 38] that parse queries into sub-queries through query graphs or iterative LLM calls, TARD learns latent intent representations that capture diverse query facets within one retrieval step. This design maintains inference efficiency while enabling diverse evidence selection.

As shown in Figure 1(b), TARD employs an adaptive intent selector to extract latent topic distributions from queries, where each dimension captures a distinct semantic perspective. The selector dynamically determines active intents per query by accumulating topic probabilities until reaching a predefined threshold. Then, an intent-aware retriever conditions relevance scoring on these intent representations. This produces intent-specific score distributions for all candidate triples. Crucially, a decorrelation loss is applied to maximize differences across the resulting score distributions, ensuring diverse triple selection. Through weak supervision defined as shortest paths between topic entities and answer entities, the retriever selects triples aligned with different query facets.

However, as shown in Figure 2, existing retrieval approaches rely on non-differentiable hard selection, which prevents gradient

propagation from the reasoner. Consequently, naive multi-intent retrieval diversification optimizes the intent selector and the retriever without generation feedback. Multiple intents often co-select the same triples (consensus). This raises two critical research questions:

- **[RQ1] How to guide intents toward reasoning performance rather than semantic relevance?** Intents optimized solely for topical similarity retrieve triples that are topically related but reasoning-irrelevant. Thus, semantic diversification alone does not guarantee improved reasoning performance.
- **[RQ2] How to appropriately utilize consensus without overemphasizing uninformative triples?** While consensus appears informative, high frequency does not ensure actual reasoning benefit. Triples with hub entities or generic relations naturally appear across multiple intents but provide minimal specific information for answering the query.

Addressing these research questions requires evaluating whether intents and consensus improve downstream generation. TARD achieves this through end-to-end optimization enabled by Gumbel-Softmax differentiable sampling, which preserves hard selection in the forward pass while allowing gradients to flow through a continuous surrogate during training. At the retriever level, supervised fine-tuning back-propagates generation feedback to jointly train the topic-adaptive intent selector and triple scorer based on reasoning performance. This shifts retrieval from semantic similarity toward reasoning-driven diversification. Diverse triple selection aligns with actual reasoning benefit and forms beneficial consensus patterns. At the generator level, adaptive direct preference optimization (DPO) [30] trains consensus utilization through reasoning performance comparison, reinforcing informative co-selected triples while suppressing uninformative high-frequency patterns.

Our main contributions are summarized as follows:

- We introduce a topic-adaptive multi-intent selector that extracts latent intents via neural topic modeling and dynamically determines active intents per query through cumulative probability thresholding, enabling diverse retrieval in one step.
- We enable end-to-end optimization via Gumbel-Softmax differentiable sampling, jointly training reasoning-driven consensus retrieval via supervised fine-tuning and consensus utilization via adaptive direct preference optimization.
- Extensive experiments on WebQSP and CWQ demonstrate that TARD significantly outperforms various types of graph-based RAG baselines, with ablation studies confirming the effectiveness of both retriever-level and generator-level optimization.

2 Related Works

2.1 Graph-based RAG for KGQA

Integrating external knowledge (e.g., KGs) in retrieval-augmented generation (RAG) has recently emerged as a key approach to improve QA performance. One line of research has proposed injecting information extracted from KGs into LLMs in the form of learnable soft prompts. G-Retriever [14] formalizes KG subgraph extraction as a Prize-Collecting Steiner Tree (PCST) optimization problem and converts the KG subgraph into a soft prompt for LLM input. GNP [45] employs a GNN-based neural prompt to inject structural

information into a frozen LLM. LightPROF [3] introduces a Knowledge Adapter, which encodes multi-hop relations into compressed prompts, while Amar [42] incorporates multi-aspect prompting with gating mechanisms to suppress noise. However, soft prompts typically collapse structural information into one or a few dense prompt embedding vectors, thereby losing fine-grained relational patterns. In contrast, we operate them at the triple level rather than the whole graph level to preserve structural information better.

Another line of research focuses on retrieving faithful substructures that are most relevant to the question. KELP [12] secured multi-hop evidence via semantic path matching. RoG [25] and GCR [26] enabled faithful reasoning by letting the LLM generate reasoning paths directly or by applying structural constraints through KG-Tries. GNN-RAG [28] propagated question embeddings over the KG to extract relevant subgraphs, while SubgraphRAG [20] achieved both efficiency and performance with MLP-based triple scoring. Despite improving retrieval diversity at the single question level, existing methods struggle to capture global consistency and diversity that reflect latent intents or topics shared across multiple questions.

2.2 End-to-End Differentiable Retrieval

Traditional RAG pipelines suffer from an optimization mismatch because the retriever and generator are trained separately. To address this issue, recent studies combine them and optimize jointly. EMDR [31] treats the retriever–reader as a latent variable and applies the Expectation–Maximization algorithm [9], showing that the retriever can be optimized by only answer-level supervision. VOD [40] proposed a variational inference–based surrogate objective, enabling efficient joint optimization in large corpus setting.

More recent work has attempted to make the retrieval step itself differentiable. D-RAG [2] applied Gumbel-Softmax sampling to allow continuous optimization of KG subgraph selection, while RAG-DDR [22] introduced differentiable data rewards that encourage the retriever’s choices to directly enhance the generator’s output quality. While these approaches improve consistency and alignment at the single question level, they still fall short in leveraging intent or topic information shared across multiple questions to ensure retrieval diversity. In this paper, we propose a novel end-to-end framework that couples intent-aware retrieval over question sets with differentiable triple selection. This framework aims to achieve both global consistency and diverse subgraph selection in KGQA.

2.3 Query Decomposition and Reformulation

Complex QA tasks deal with complex queries that require multi-hop reasoning and multiple semantic aspects. Single-pass retrieval often fails to address such complexity [5, 33]. Recent works decompose or reformulate the original query to directly reduce their complexity. On the one hand, complex queries are decomposed into sub-queries via graph-structured intermediates that encode compositionality [15, 17, 23] or mapped to subgraph fragments via fine-grained phrase-level semantic parsing [35]. These approaches then perform retrieval and reasoning sequentially at each step. On the other hand, some methods progressively reformulate the original query through multiple rounds of LLM-KG interaction [19, 38],

sequentially accumulating evidence through iterative read and reason cycles or iterative graph traversal over structured knowledge.

Instead of multiple LLM calls for explicit decomposition or reformulation, our work performs single-pass retrieval while learning latent intents that simultaneously select evidence across multiple semantics, achieving diverse evidence and efficient inference.

3 Task Formulation

Let a knowledge graph (KG) be defined as $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$, where $\mathcal{E}$ and $\mathcal{R}$ denote the sets of entities and relations, and $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$ denotes the set of triples. Each triple $(h, r, t) \in \mathcal{T}$ represents a factual relation from a head entity h to a tail entity t .

Given a natural language question q , the task of KGQA is to identify the correct answer entities by reasoning over multiple related triples in the knowledge graph. Under a graph-based retrieval-augmented generation framework, KGQA is typically addressed in two stages. First, a retriever selects a question-relevant subset of triples $\mathcal{T}(q) \subseteq \mathcal{T}$, which forms a subgraph containing information useful for answering the question. Second, the selected subgraph is provided to a large language model (LLM), which performs reasoning over the retrieved evidence and generates the final answer.

4 The Proposed Method

We present TARD, a multi-intent retrieval framework for KGQA that models latent intents. TARD extracts multiple intents from a question and performs intent-conditioned retrieval over the knowledge graph. Retrieved triples are aggregated via a consensus mechanism to form a unified prompt. To align retrieval with generation, TARD uses Gumbel-Softmax for differentiable triple selection. We further apply Adaptive DPO to guide consensus utilization. The overall architecture of TARD is illustrated in Figure 3.

4.1 Topic-Adaptive Intent Selector

Existing KGQA retrievers often compress each question into a single embedding, which can be insufficient for multi-faceted queries. To capture diverse retrieval perspectives, we represent each query as a distribution over K latent intents shared across the query collection.

Given a query q , let $\mathbf{q} \in \mathbb{R}^D$ be its embedding. We first obtain a hyperspherical latent representation by applying an encoder $f_{\text{enc}}(\cdot)$ and ℓ_2 normalization:

$$\mathbf{z} = \frac{f_{\text{enc}}(\mathbf{q})}{\|f_{\text{enc}}(\mathbf{q})\|_2}. \quad (1)$$

Based on $\mathbf{z}$, we infer a query-conditioned intent distribution over K global intents:

$$\boldsymbol{\theta} = \text{softmax}(\mathbf{W}\mathbf{z}), \quad \boldsymbol{\theta} \in \mathbb{R}^K, \quad (2)$$

where $\mathbf{W} \in \mathbb{R}^{K \times D}$ is a learnable projection matrix, and θ_k denotes the probability of intent k for the query. Each intent k is represented by a learnable intent embedding $\mathbf{u}_k \in \mathbb{R}^D$ shared across all queries. The query-specific distribution $\boldsymbol{\theta}$ assigns probability to multiple intents, allowing a single query to express multiple retrieval perspectives.

At inference time, we activate intents adaptively since different queries may require different numbers of retrieval perspectives. Specifically, we sort intents in descending order of θ_k and accumulate them until the cumulative probability exceeds a threshold γ ,

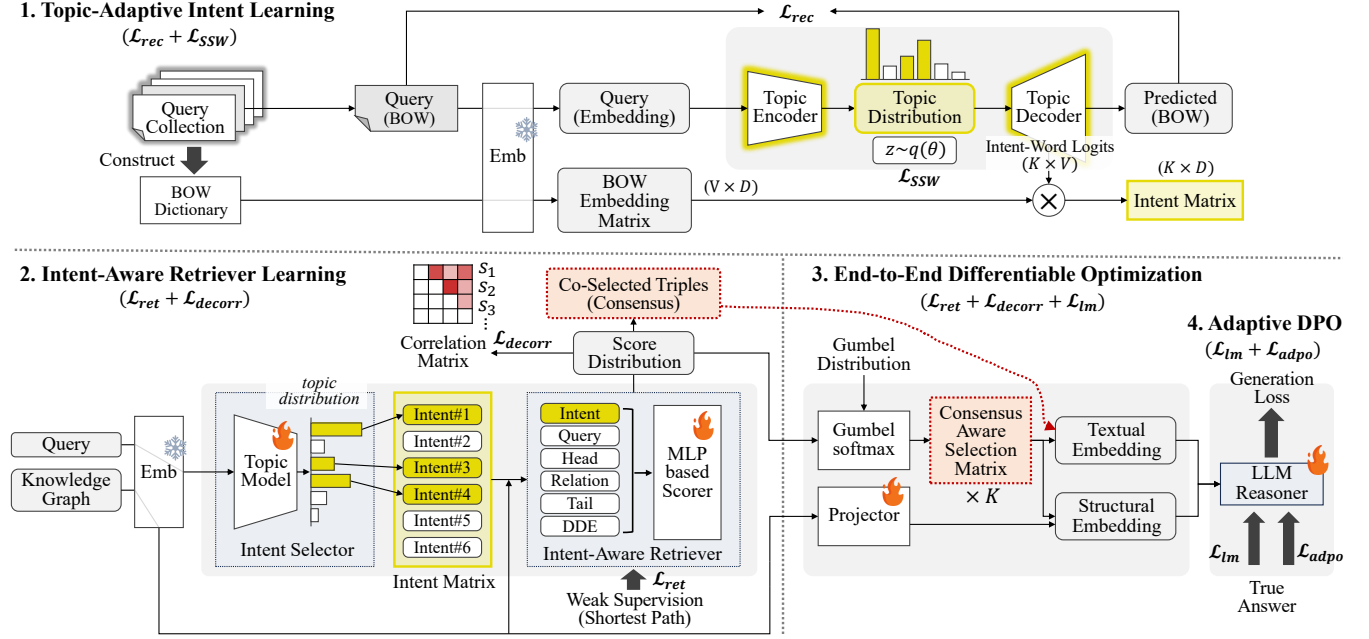


Figure 3: TARD’s four-stage training framework. Stage 1 (top): Topic-adaptive intent learning trains an encoder-decoder topic model ($\mathcal{L}_{rec} + \mathcal{L}_{SSW}$) to extract latent intents, producing an intent matrix. Stage 2 (bottom-left): Intent-aware retriever learning uses an intent selector and triple scorer to generate intent-specific score distributions; decorrelation loss ($\mathcal{L}_{decorr}$) diversifies selection to form consensus across intents, with shortest path supervision ($\mathcal{L}_{ret}$). Stage 3 (bottom-right): End-to-end differentiable optimization via Gumbel Softmax enables gradient-based training of the pipeline through supervised fine-tuning ($\mathcal{L}_{lm}$). This weights consensus triples and aligns retrieval-generation with reasoning performance. Stage 4: Adaptive DPO ($\mathcal{L}_{adpo}$) context-aware consensus use from reasoning feedback, encouraging effective consensus use and favoring informative evidence.

forming the active intent set $\mathcal{I}(q)$:

$$\sum_{k \in \mathcal{I}(q)} \theta_k \geq \gamma, \quad k_{\min} \leq |\mathcal{I}(q)| \leq k_{\max}. \quad (3)$$

Here, $k_{\min}$ ensures at least one intent is always selected, while $k_{\max}$ prevents excessive activation to keep retrieval cost bounded. As illustrated in Fig. 5, the activated intent subsets vary across queries, reflecting query-adaptive multi-intent selection.

Following S2WTM [1], we train the intent selector using a reconstruction objective and spherical sliced-Wasserstein (SSW) regularization. The reconstruction loss encourages the inferred intent distribution to preserve lexical semantics by reconstructing a bag-of-words (BoW) target:

$$\mathcal{L}_{rec} = \mathbb{E}_{(q,t)} \left[- \sum_{v=1}^V t_v \log \hat{t}_v(\theta(q)) \right], \quad (4)$$

where $t \in \mathbb{R}^V$ is the observed BoW representation and $\hat{t}(\theta(q)) \in \mathbb{R}^V$ denotes the decoder output conditioned on the intent distribution.

To regularize the hyperspherical latent space, SSW aligns the aggregated posterior of latent representations with a hyperspherical prior:

$$\mathcal{L}_{SSW} = \text{SSW}^2(q(z), p(z)), \quad (5)$$

which stabilizes the latent geometry and prevents degenerate solutions in which intent representations collapse to a narrow region of the hypersphere.

4.2 Intent-Aware Retriever

The intent-aware retriever scores triple relevance conditioned on the activated intent set $\mathcal{I}(q)$. For each intent $k \in \mathcal{I}(q)$, it independently computes relevance scores for candidate triples $(h, r, t) \in \mathcal{T}(q)$ as

$$s_k(q, h, r, t) = f_{\text{pred}}([u_k \| q \| h \| r \| t \| \phi_{\text{DDE}}]), \quad (6)$$

where h , r , and t denote the head, relation, and tail embeddings, and ϕ_{DDE} represents structural features.

Without additional constraints, multiple intents may converge on the same triples, leading to redundant evidence selection. To mitigate this issue, we introduce an intent-wise decorrelation regularizer over score distributions. For each intent k , we normalize its score vector into a probability distribution $\mathbf{P}_k = \text{softmax}(\mathbf{s}_k)$ and penalize correlations across intents:

$$\mathcal{L}_{decorr} = \frac{1}{|\mathcal{I}(q)|(|\mathcal{I}(q)| - 1)} \sum_{k \neq k'} |\text{Corr}(\mathbf{P}_k, \mathbf{P}_{k'})|, \quad (7)$$

where $\text{Corr}(\cdot, \cdot)$ denotes the Pearson correlation coefficient. This regularization encourages different intents to prioritize complementary subsets of triples rather than repeatedly selecting the same

evidence. The retriever is trained using a binary cross-entropy (BCE) objective for intent-conditioned triple relevance supervision, augmented with the decorrelation regularizer.

4.3 Differentiable Triple Selection

Since triple selection is discrete operation, gradients cannot flow through the selection step, making end-to-end training difficult. We address this using a Gumbel-Softmax reparameterization with a straight-through estimator (STE) [2, 18].

For each query q and intent $k \in \mathcal{I}(q)$, the retriever produces an intent-conditioned relevance score $s_{f,k}$ for each candidate triple $f \in \mathcal{T}(q)$ (Section 4.2). We interpret $s_{f,k}$ as a Bernoulli logit for triple selection under intent k and obtain the selection probability $z_{f,k} = \sigma(s_{f,k})$.

To enable differentiable sampling, we apply a binary Gumbel-Softmax relaxation:

$$z_{f,k}^{\text{soft}} = \text{softmax} \left(\begin{pmatrix} (\log z_{f,k} + \eta_{f,k}^{(1)})/\tau_g \\ (\log(1 - z_{f,k}) + \eta_{f,k}^{(2)})/\tau_g \end{pmatrix} \right), \quad (8)$$

where $\eta_{f,k}^{(1)}, \eta_{f,k}^{(2)} \sim \text{Gumbel}(0, 1)$ and τ_g denotes the temperature parameter. We use the first component of $z_{f,k}^{\text{soft}}$ as the soft selection score,

$$m_{f,k}^{\text{soft}} = z_{f,k}^{\text{soft}}[0]. \quad (9)$$

To enforce a fixed retrieval budget and keep inference cost bounded, we perform a per-intent Top- B selection based on $m_{f,k}^{\text{soft}}$, yielding a hard binary mask:

$$m_{f,k}^{\text{hard}} = \mathbb{I}[f \in \text{TopB}(\{m_{f',k}^{\text{soft}} \mid f' \in \mathcal{T}(q)\})]. \quad (10)$$

Finally, we apply the straight-through estimator:

$$m_{f,k} = m_{f,k}^{\text{hard}} + m_{f,k}^{\text{soft}} - \text{SG}(m_{f,k}^{\text{soft}}), \quad (11)$$

where $\text{SG}(\cdot)$ denotes the stop-gradient operator.

4.4 Consensus-aware Prompt Construction

To enable end-to-end optimization with generation feedback, we construct an embedding-level prompt that preserves differentiability throughout the selection process. As described in Section 4.3, an STE mask $m_{f,k} \in [0, 1]$ is obtained for each triple-intent pair, following hard Top- B selection in the forward pass.

Intent-wise selections are aggregated to compute a per-triple consensus strength:

$$g_f = 1 + \frac{1}{|\mathcal{I}(q)|} \sum_{k \in \mathcal{I}(q)} m_{f,k}. \quad (12)$$

With STE, $g_f > 1$ in the forward pass if and only if triple f is selected by at least one active intent, and larger values indicate stronger cross-intent support. The consensus score g_f is therefore used both as a selection signal ($g_f > 1$) and as a soft weighting factor that emphasizes triples with higher agreement. Lightweight tokenized textual markers indicating consensus levels are included as auxiliary structural cues, while the main conditioning signal is provided by the embedding-level prompt.

Each triple $f = (h, r, t)$ is encoded from two complementary views. The structural view captures graph-level information by

projecting pretrained entity and relation embeddings into the LLM embedding space via an MLP:

$$\mathbf{e}_{\text{struct}} = \text{MLP}([\mathbf{h} \parallel \mathbf{r} \parallel \mathbf{t}]).$$

The semantic view encodes the textualized triple using the LLM input embedding layer:

$$\mathbf{e}_{\text{sem}} = E_{\text{LLM}}(\text{Tokenize}(h, r, t)).$$

Both views are scaled by the consensus score g_f and concatenated:

$$\widehat{\mathbf{B}}(f) = [g_f \cdot \mathbf{e}_{\text{struct}}; g_f \cdot \mathbf{e}_{\text{sem}}]. \quad (13)$$

In addition, a lightweight consensus-dependent bias is injected into the structural embedding, allowing consensus to influence not only the strength but also representation direction.

Finally, the prompt is formed by concatenating embeddings of selected triples in descending order of g_f :

$$\mathcal{P}(q) = \text{Concat}(\{\widehat{\mathbf{B}}(f) \mid f \in \mathcal{T}(q), g_f > 1\}). \quad (14)$$

4.5 Adaptive Preference Optimization

Emphasizing consensus across multiple intents can improve generation quality, but consensus is not always beneficial. To adaptively regulate its influence, we introduce a preference optimization objective that learns when consensus should be encouraged and when it should be suppressed, while preserving robust behavior under plain retrieval contexts. See Appendix E for case studies where consensus is beneficial or harmful.

For each query q , we construct two retrieval-augmented prompts: a consensus-weighted prompt $C^w(q)$ and a plain prompt $C^p(q)$ derived from the same retrieved triples without consensus weighting. The model generates two responses, y^w and y^p , enabling a direct comparison between weighted and plain contexts.

We define the conditional log-likelihood score of a response y under context C as:

$$s_\theta(y; q, C) = \log \pi_\theta(y \mid q, C). \quad (15)$$

Then, their relative preference margin is computed as:

$$\Delta(q) = s_\theta(y^w; q, C^w(q)) - s_\theta(y^p; q, C^p(q)). \quad (16)$$

Here, $C^w(q)$ is the prompt constructed with consensus-weighted triples and $C^p(q)$ is the prompt constructed from the same retrieval results without consensus weighting (i.e., plain context).

A preference is assigned only when a non-trivial margin exists:

$$\text{Pref}(q) = \begin{cases} y^w, & \Delta(q) > \delta, \\ y^p, & \Delta(q) < -\delta, \\ \text{None}, & \text{otherwise.} \end{cases} \quad (17)$$

and optimization is applied selectively:

$$\mathcal{L}_{\text{adpo}}(q) = \begin{cases} -\log \sigma(\beta \Delta(q)), & \Delta(q) > \delta, \\ -\log \sigma(-\beta \Delta(q)), & \Delta(q) < -\delta, \\ 0, & \text{otherwise.} \end{cases} \quad (18)$$

Here, β is a scaling coefficient. This objective reinforces consensus weighting when it clearly improves generation, suppresses it when harmful, and ignores uncertain cases, allowing the model to adaptively use consensus while preserving robust behavior under plain retrieval context

4.6 Training Strategy and Inference

We use a four-stage training procedure to organize the learning process and define the role of each component.

Stage 1 / 2: Pretraining. We pretrain the intent selector and the retriever separately to obtain stable initialization before end-to-end optimization. The intent selector is optimized using the reconstruction loss and spherical sliced-Wasserstein regularization defined in Section 4.1,

$$\mathcal{L}_{\text{sel}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{SSW}}, \quad (19)$$

which encourages meaningful intent representations while preventing degenerate latent solutions.

The retriever is pretrained to estimate triple-level relevance under each activated intent, establishing a meaningful retrieval prior before optimization with generation feedback. During this stage, it is optimized using the retrieval loss $\mathcal{L}_{\text{ret}}$ together with the decorrelation regularizer $\mathcal{L}_{\text{decorr}}$, which promotes diversity across intent-specific retrievals.

Stage 3: Joint End-to-End Training. Using the pretrained selector and retriever as initialization, we jointly fine-tune the retriever and language model end-to-end. Gradients are propagated to the retriever through differentiable prompt construction. We optimize the following objective:

$$\mathcal{L}_{\text{joint}} = \rho \mathcal{L}_{\text{ret}} + (1 - \rho) \mathcal{L}_{\text{lm}} + \rho_{\text{decorr}} \mathcal{L}_{\text{decorr}}, \quad (20)$$

where $\rho \in [0, 1]$ balances retrieval learning and language modeling, and $\rho_{\text{decorr}} \in [0, 1]$ controls the strength of the decorrelation regularizer.

Stage 4: Adaptive Preference Optimization. We freeze the selector and retriever and fine-tune only the language model using the ADPO objective defined in Sec. 4.5 (Eqs. 16–18). For each query q , we construct $C^w(q)$ and $C^p(q)$ from the same retrieval results and apply updates only when the preference margin exceeds δ . The overall objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{lm}} + \rho_{\text{adpo}} \mathcal{L}_{\text{adpo}}. \quad (21)$$

where ρ_{adpo} is a hyperparameter that weights the ADPO preference optimization term.

Inference. We first obtain $\mathcal{I}(q)$ from the intent selector (Section 4.1). The retriever selects Top- B triples per active intent and merges them into an evidence subgraph, which is converted into a prompt for the language model to generate the final answer.

5 Experiment

In this section, we conduct a comprehensive empirical evaluation of our proposed framework, TARD. Our experiments are designed to systematically address the following research questions (RQs):

- **RQ1:** Does TARD outperform existing KGQA baselines on benchmark datasets, and how consistently does it improve across metrics?
- **RQ2:** How does each core component contribute to the model’s overall performance?
- **RQ3:** How does multi-intent retrieval affect retrieval coverage and evidence complementarity compared to existing approaches?
- **RQ4:** Does adaptive consensus validation mitigate hub-node amplification while reinforcing reasoning-relevant evidence?

Table 1: Statistics of datasets.

Datasets	#Train	#Test	Max #hop
WebQSP	2,826	1,628	2
CWQ	27,639	3,531	4

5.1 Experimental Setup

5.1.1 Datasets. The experimental evaluation was conducted on two widely used real-world KGQA benchmarks: WebQSP [44] and ComplexWebQuestions (CWQ) [39]. Both WebQSP and CWQ are grounded in the Freebase knowledge graph [6], which contains approximately 86M entities and 4,218 relations. WebQSP consists of natural language questions paired with logical forms and primarily focuses on single-entity questions with relatively simple reasoning structures. In contrast, CWQ is a more challenging benchmark constructed by automatically generating complex questions from WebQuestions, emphasizing multi-hop and compositional reasoning. In addition to these real-world benchmarks, we conduct supplementary experiments on MetaQA-3hop [48] to further examine the effectiveness of our approach in explicit multi-hop reasoning scenarios. MetaQA is a synthetic benchmark designed with fixed hop lengths, enabling clear evaluation of multi-hop reasoning performance. Detailed experimental results on MetaQA are reported in Appendix A.

Dataset statistics for the main benchmarks are summarized in Table 1.

5.1.2 Baselines. We compare TARD with representative KGQA baselines across three categories: (i) Embedding-based (ii) Graph Reasoning-based (iii) Graph-augmented LLM-based. Embedding-based KGQA methods compress each question into a single representation and perform multi-hop reasoning within a shared embedding space. EmbedKGQA [32] learns entity and relation embeddings for KB completion, TransferNet [34] transfers reasoning patterns across domains, and ReTraCK [8] retrieves and tracks entities through relation chains. For Graph Reasoning without LLM methods explicitly exploit knowledge graph structure. GraftNet [37] and PullNet [36] perform message passing over query-specific subgraphs. NSM [13] conducts step-wise neural symbolic reasoning, SR+NSM [46] augments it with trainable subgraph retrieval, and ReaRev [27] iteratively refines reasoning paths. For Graph+LLM (GraphRAG) methods combine graph-based retrieval with LLM reasoning. RoG [25] generates relation paths within a planning-retrieval-reasoning framework. G-Retriever [14] uses cosine similarity search to extract relevant subgraphs. GNN-RAG [28] predicts answer entities with lightweight GNNs and extracts supporting paths. SubgraphRAG [20] leverages directional structural encoding for efficient subgraph retrieval.

5.1.3 Evaluation Metrics. We use Hit and Macro F1 as evaluation metrics. **Hit** evaluates whether the predicted candidate set contains at least one correct answer, regardless of its rank, which captures the model’s ability to successfully retrieve relevant entities even if they are not placed at the top. **Macro F1** accounts for the fact that a question may correspond to multiple correct answers by balancing

Table 2: Performance comparison with baselines on WebQSP and CWQ. "↔" denotes cross-dataset evaluation (trained on one dataset, tested on the other).

Type	Method	WebQSP		CWQ	
		Hit	F1	Hit	F1
Embedding	EmbedKGQA [32]	66.6	66.6	34.5	33.4
	TransferNet [34]	71.4	48.6	37.2	36.8
	ReTraCk [8]	69.8	50.1	39.6	38.2
Graph Reasoning	GraftNet [37]	66.7	62.8	36.8	32.1
	PullNet [36]	68.1	65.2	45.9	41.3
	NSM [13]	68.7	46.7	42.4	-
	SR+NSM [46]	69.5	50.2	47.1	43.7
	ReaRev [27]	76.4	70.9	52.9	-
Graph+LLM (GraphRAG)	ToG (GPT-4) [38]	82.6	69.5	61.2	54.8
	RoG [25]	85.7	70.8	62.6	56.2
	G-Retriever [14]	70.1	64.3	48.2	42.7
	GNN-RAG [28]	85.7	71.3	66.8	59.4
	SubgraphRAG [20]	86.6	70.6	57.0	47.2
	SubgraphRAG (↔) [20]	83.4	66.4	48.5	37.96
Our	TARD	88.3	75.8	73.7	62.3
	TARD (↔)	85.3	73.6	61.2	52.8

precision and recall, providing a comprehensive assessment of both the correctness and coverage of the predicted answers.

5.1.4 Implementations. TARD is composed of a topic modeling based intent selector and a lightweight multi-intent retriever that operates at the triple level. For text representation, we use GTE-large-en-v1.5 [24] as a pre-trained text encoder to obtain dense embeddings of textual inputs used throughout the retrieval process. As the reasoning backbone, we adopt Llama-3.1-8B-Instruct [10]¹. To enable efficient experimentation, the model is fine-tuned using LoRA with 4-bit quantization. All experiments are conducted on a single NVIDIA A100 80GB GPU. For fair comparison across methods, we restrict all experiments to the top 100 retrieved triples. Detailed hyperparameter settings are provided in Appendix B.

5.2 Overall Performance (RQ1)

Table 2 shows that TARD consistently outperforms all baselines across both benchmark datasets. On WebQSP, TARD achieves 88.3% Hit and 75.8% F1, surpassing all existing baselines. On CWQ, TARD achieves 73.7% Hit and 62.3% F1, outperforming the previous best result from GNN-RAG [28]. This is particularly notable given that CWQ emphasizes complex multi-hop reasoning. The substantial improvements on CWQ demonstrate that TARD’s multi-intent retrieval is especially effective for queries requiring complex reasoning rather than simple queries.

Cross-dataset evaluation (denoted as "↔" in Table 2) demonstrates TARD’s strong generalization capability. When trained on one dataset and evaluated on another, TARD maintains consistent performance across all settings. Specifically, training on WebQSP and testing on CWQ yields 61.2% Hit and 52.8% F1, while training

Table 3: Ablation study on WebQSP and CWQ.

Variant	WebQSP		CWQ	
	Hit	F1	Hit	F1
TARD (Full)	88.3	75.8	73.7	62.3
w/o Gumbel-softmax	86.1	69.9	67.3	57.8
w/o Intent Selector	86.5	71.3	70.3	59.7
w/o Consensus Weighting	87.5	70.6	69.1	57.6
w/o Decorrelation	86.8	69.6	69.5	57
w/o Adaptive DPO	87.9	75.3	70.7	61.3

on CWQ and testing on WebQSP achieves 85.3% Hit and 73.6% F1. This indicates that TARD’s multi-intent approach generalizes robustly across reasoning without overfitting to dataset-specific patterns.

These results highlight two key observations: (i) multi-intent retrieval, which interprets queries from multiple perspectives, is more effective than conventional approaches that compress queries into single embeddings, and (ii) explicit diversity enforcement through decorrelation becomes more critical for queries requiring complex reasoning paths. The large improvements on CWQ suggest that different intents interpret the same query from distinct perspectives, exploring different reasoning paths to effectively gather complementary evidence. This enables the model to capture important reasoning clues that single-embedding approaches might miss, ultimately providing richer and more diverse context to the language model for more accurate answer generation.

5.3 Ablation Study (RQ2)

We evaluate the contribution of each component through ablation analysis on the WebQSP and CWQ test sets. As shown in Table 3, each component provides distinct benefits to the overall framework. Removing Gumbel-Softmax results in the largest performance drop because it blocks gradient flow from generation to retrieval. Without differentiable selection, the retriever cannot receive feedback from the generation outcome and relies solely on supervised labels. The Adaptive Intent Selector and Decorrelation Loss show comparable impact, working synergistically to enable multi-intent diversification. The selector generates diverse retrieval perspectives, while decorrelation enforces that these perspectives select different evidence. Without the selector, the system regresses to single-embedding retrieval; without decorrelation, multiple intents collapse onto the same triples. Removing Consensus Weighting or Adaptive DPO results in smaller but consistent drops across both datasets. Consensus Weighting provides cross-intent agreement signals to identify reliable evidence, while Adaptive DPO refines how these signals are utilized during generation. Both components contribute to consensus-based reasoning but at different stages of the pipeline.

5.4 Further Analysis (RQ3, RQ4)

5.4.1 Visualization of adaptive intent selection. To qualitatively verify that the intent selector behaves adaptively across queries, we visualize the predicted intent selection probabilities for a small

¹<https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct>

set of example questions, which can be found in the Appendix 5. As shown in the heatmap, probability mass is concentrated on a small subset of intents for each question, while the activated intents vary across questions. This observation supports our design that different queries can trigger different numbers and types of intents, providing diverse retrieval perspectives when needed.

5.4.2 Intent-Level Retrieval Diversity. To assess whether different intents produce genuinely diverse retrieval, we measure the mean pairwise Jaccard overlap between intent-specific top- B triple sets. Lower overlap indicates that different intents retrieve distinct evidence rather than redundant triples. Table 4 shows that TARD achieves low Jaccard overlap (0.28 ~ 0.40) across both datasets. This indicates that different intents retrieve distinct triple sets, with approximately 70% of triples being non-overlapping. As budget increases, overlap rises moderately (e.g., 0.28→0.37 on WebQSP), which is expected—larger budgets naturally lead each intent to include commonly important triples.

5.4.3 Coverage Preservation under Diversification. To verify that diverse retrieval does not miss essential evidence for reaching answers, we measure Shortest-Path Recall@B, which indicates how many triples on gold-standard reasoning paths are successfully retrieved. Table 4 demonstrates that TARD maintains essential coverage while retrieving diverse evidence. On WebQSP, the recall gap compared to the baseline exists at smaller budgets ($\Delta\text{Recall}=-0.087$ at $B=30$) but nearly vanishes as budget increases (-0.016 at $B=100$). More notably, on CWQ, TARD achieves higher recall than the baseline ($+0.011$ at $B=50$, $+0.013$ at $B=100$). This indicates that diverse retrieval goes beyond merely not sacrificing coverage, it actually discovers complementary reasoning paths that conventional approaches miss.

5.4.4 Adaptive Training Dynamics. To verify selective consensus utilization, we analyze training patterns on 2,708 samples from the WebQSP dataset. Weighted contexts outperform plain contexts in 11.7% of cases and underperform in 4.1%, while the majority of cases (84.2%) show comparable performance (Figure 4a). This indicates that consensus benefits are selective rather than universal.: weighted contexts help approximately three times more often than they harm, but in most cases, they neither significantly help nor harm. These findings justify adaptive reinforcement of consensus rather than applying it universally, as blanket emphasis would introduce noise in the majority of cases where consensus provides minimal discriminative value.

5.4.5 Consensus Utilization Improvement. We measure consensus utilization effectiveness through the log-likelihood advantage $\Delta(q)$ (Eq. 16) on the test set. After Adaptive DPO, this advantage nearly doubles, increasing from $\Delta(q) = 0.079$ to $\Delta(q) = 0.148$ (+87.8%), with the rate of beneficial cases ($\Delta(q) > 0$) increasing from 62.1% to 63.9%. To examine how consensus strength affects utilization, we stratify queries by their maximum consensus score—the highest number of intents co-selecting the same triple. Figure 4b shows that the improvement scales with consensus strength: queries with maximum scores of 2, 3, and 4 show $\Delta(q)$ gains of +0.047, +0.068, and +0.071, respectively. This progressive increase confirms that ADPO effectively learns to leverage stronger consensus signals, with higher intent agreement leading to greater utilization benefits.

Table 4: Retrieval diversity and coverage trade-off

Dataset	Budget	Jaccard↓	Shortest-Path Recall@B		ΔRecall
			Single	TARD	
WebQSP	30	0.28±0.16	0.775±0.342	0.688±0.373	−0.087
	50	0.32±0.16	0.824±0.308	0.781±0.329	−0.043
	100	0.37±0.17	0.883±0.251	0.867±0.262	−0.016
CWQ	30	0.30±0.19	0.639±0.357	0.631±0.354	−0.008
	50	0.35±0.20	0.711±0.334	0.722±0.327	+0.011
	100	0.40±0.20	0.796±0.291	0.808±0.278	+0.013

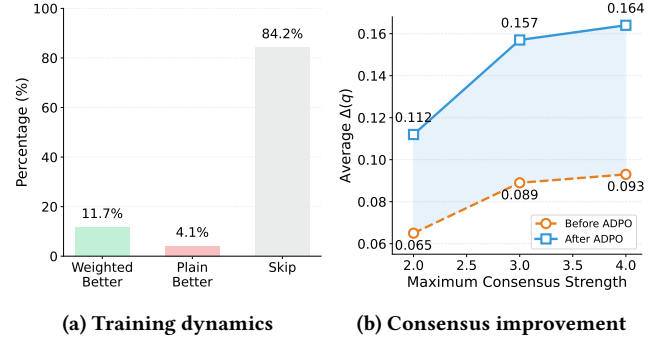


Figure 4: Adaptive DPO analysis. (a) Training dynamics. (b) $\Delta(q)$ gains increase with consensus strength.

6 Conclusion

We introduced TARD, a topic-adaptive retrieval diversification framework for Knowledge Graph Question Answering (KGQA). Unlike existing graph-based RAG approaches that assume a single dominant intent, TARD learns query-conditioned intent distributions to retrieve complementary evidence for multi-hop reasoning, combining intent diversification, consensus-based reliability, and differentiable optimization. Experiments on WebQSP and CWQ demonstrate that TARD achieves state-of-the-art performance across both benchmarks. Results highlight the importance of reasoning-driven diversification in improving overall QA performance. Future work includes extending TARD to hybrid retrieval scenarios combining knowledge graphs with unstructured text and applying intent-adaptive diversification to other reasoning-intensive tasks such as fact verification and multi-modal QA. Our findings suggest that modeling latent multi-intent structure is a promising direction for aligning retrieval with downstream reasoning objectives in retrieval-augmented systems.

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GENAI USAGE DISCLOSURE

We only used generative AI tools for grammar checking and sentence refinement. This paper does not include any content fully generated by large language models. All ideas and experimental analyses were conceived and conducted solely by the authors.

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A Additional Results on MetaQA

We further evaluate our method on MetaQA-3hop [48], a synthetic KGQA benchmark with fixed three-hop reasoning. MetaQA-3hop contains 114,186 training and 14,274 test questions, providing explicit multi-hop reasoning paths. This setting allows us to examine whether TARD remains effective when longer reasoning chains are required.

Table 5 reports Hit performance. TARD achieves the best results among compared methods, indicating that multi-intent retrieval and consensus-aware evidence utilization remain effective under fixed multi-hop reasoning.

Table 5: Hit performance on MetaQA-3hop.

Method	Hit (%)
RoG	84.0
SubgraphRAG	76.0
TARD (ours)	88.7

B Implementation Details

The intent selector employs an adaptive cumulative threshold γ to determine the number of active intents per query, with dataset-specific bounds ($k_{\min}, k_{\max}$) to avoid degenerate cases. To encourage diversity among intent-specific retrievals, we apply a decorrelation weight ρ_{decorr} .

Joint training is conducted for 10 epochs on WebQSP and 15 epochs on CWQ. The overall objective balances retrieval supervision and generation loss using a mixing coefficient ρ , while each component is optimized with its own learning rate. The LLM backbone is fine-tuned using LoRA.

Finally, we apply Adaptive Preference Optimization (ADPO) for a small number of epochs to selectively reinforce consensus-based conditioning only when it yields clear generation improvements. All hyperparameters used in our experiments are summarized in Table 6.

Table 6: Hyperparameters for WebQSP and CWQ.

Module	Hyperparameter	Dataset	
		WebQSP	CWQ
Intent Selector	Cumulative threshold γ	0.4	0.3
	Intent bounds ($k_{\min}, k_{\max}$)	(1,4)	(1,5)
Retriever	Decorrelation weight ρ_{decorr}	0.4	0.4
Joint Training	Intent selector LR	$1e-6$	$1e-6$
	Retriever LR	$3e-5$	$3e-5$
	LLM LR	$1e-5$	$1e-5$
	Loss balancing weight ρ	0.5	0.5
	temperature of Gumbel-softmax τ_g	0.5	0.5
Adaptive DPO	LLM LR	$1e-6$	$1e-6$
	ADPO weight ρ_{adpo}	0.1	0.1
	Optimization epochs	2	2
	Preference threshold δ	0.02	0.02

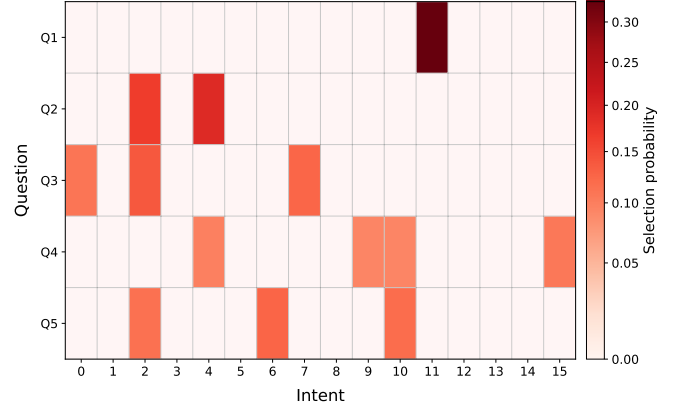


Figure 5: Heatmap visualization of intent selection probabilities (darker indicates higher probability).

C Visualization of Intent Activations

Figure 5 visualizes intent selection probabilities for a few example questions.

D Additional Experiments

D.1 Loss Balance Weight (ρ)

Figure 6 shows the impact of ρ on performance. Overall, the system demonstrates stable performance across a wide range of ρ values, indicating that the framework is robust to this hyperparameter and does not appear to require precise tuning.

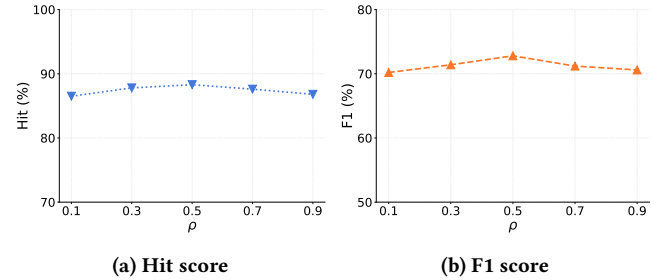


Figure 6: Sensitivity analysis of loss balance weight ρ on WebQSP.

D.2 Preference Margin (δ)

During Adaptive Preference Optimization, we gate preference updates using a margin threshold δ (Section 4.5). Figure 7 shows the sensitivity of δ on WebQSP. Overall, a moderate margin performs best: compared to the *No DPO* baseline (87.90 Hit), setting $\delta = 0$ slightly improves performance to 88.02, and $\delta = 0.02$ yields the best result (88.30). However, a larger margin $\delta = 0.05$ degrades performance to 87.80, suggesting that overly conservative gating reduces the benefit. We therefore use $\delta = 0.02$ as the default setting.

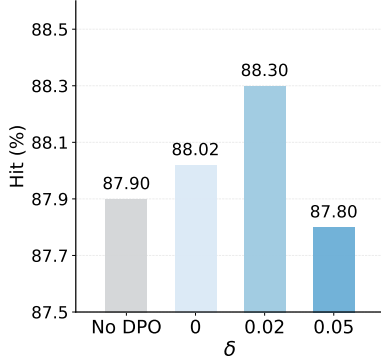


Figure 7: Sensitivity analysis of preference margin δ on WebQSP.

Stage	/epoch	Total
Intent Selector (100 ep)	10s	~17min
Retriever (15 ep)	1m	~15min
Joint Training (10 ep)	45m	~7.5h
ADPO (1 epoch)	40m	~1h

Table 7: Approx. runtime on WebQSP.

D.3 Runtime and Computational Cost (WebQSP)

We report wall-clock runtime on WebQSP, where one epoch corresponds to 2,826 training instances. Table 7 summarizes the per-epoch time and total time for each training stage. Retriever and intent selector pretraining require little wall-clock time since both components are lightweight. In contrast, joint training is substantially more time-consuming, taking about 45 minutes per epoch, as the computation is dominated by the LLM generation stage, which results in a longer overall training time.

E Case Studies: Transition Patterns of Consensus Utilization

Figure 8 presents two representative transition cases that illustrate how ADPO changes the generator’s behavior under consensus-weighted evidence. For each query, we list the question and the high-confidence triples that are strongly co-selected across multiple intents (i.e., high consensus strength). These cases highlight two complementary effects: ADPO can (i) reinforce informative consensus to improve answer coverage, and (ii) suppress uninformative high-frequency consensus to avoid generic amplification.

E.1 Case 1 (Afrikaans, WebQTest-1758): Reinforcing informative consensus.

This query requires broad geographic coverage. The listed high-consensus triples consistently support that Afrikaans is spoken in multiple countries (e.g., Namibia, South Africa, Botswana, Zambia, Malawi). Before ADPO, the model under-utilizes these informative

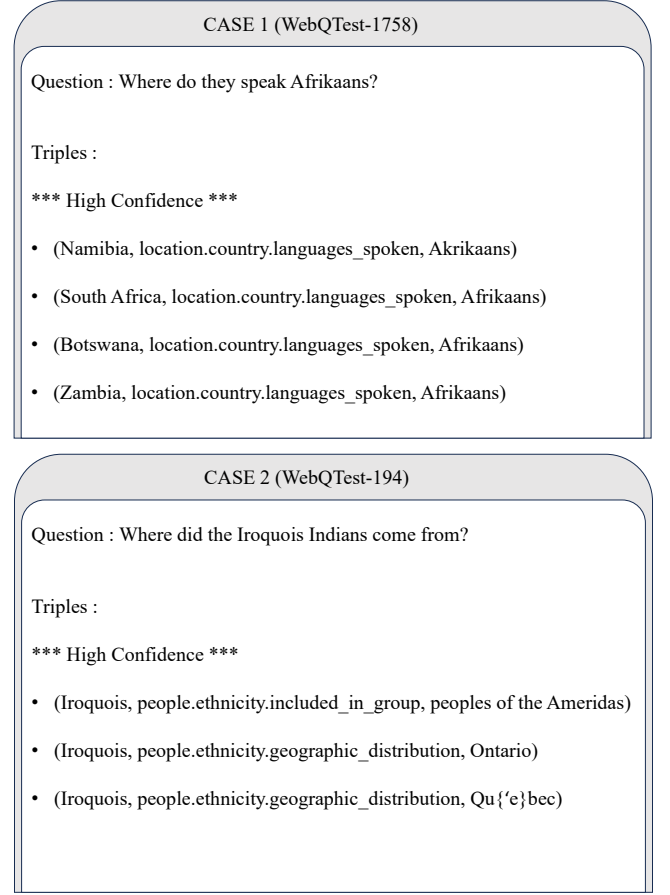


Figure 8: Case studies illustrating ADPO transition patterns. Case 1 reinforces informative consensus to expand answer coverage, while Case 2 suppresses uninformative consensus and prioritizes specific evidence.

co-selected facts under the weighted context, resulting in under-generation (missing many gold answers). After ADPO, the generator increases its reliance on such informative consensus evidence, expanding answer coverage and correcting the under-generation failure mode.

E.2 Case 2 (Iroquois, WebQTest-194): Suppressing uninformative consensus.

This query requires specific geographic origins. The baseline model tends to amplify a generic hub-like relation that is frequently co-selected but not directly helpful (e.g., *included_in_group* → *Indigenous peoples of the Americas*), which leads to overly generic outputs under the weighted context. After ADPO, the model down-weights these uninformative high-frequency patterns and prioritizes more specific location evidence (e.g., *geographic_distribution* → Ontario/Québec), producing correct and grounded answers. Overall, these transitions support our claim that ADPO learns when consensus is informative and when it should be discounted.